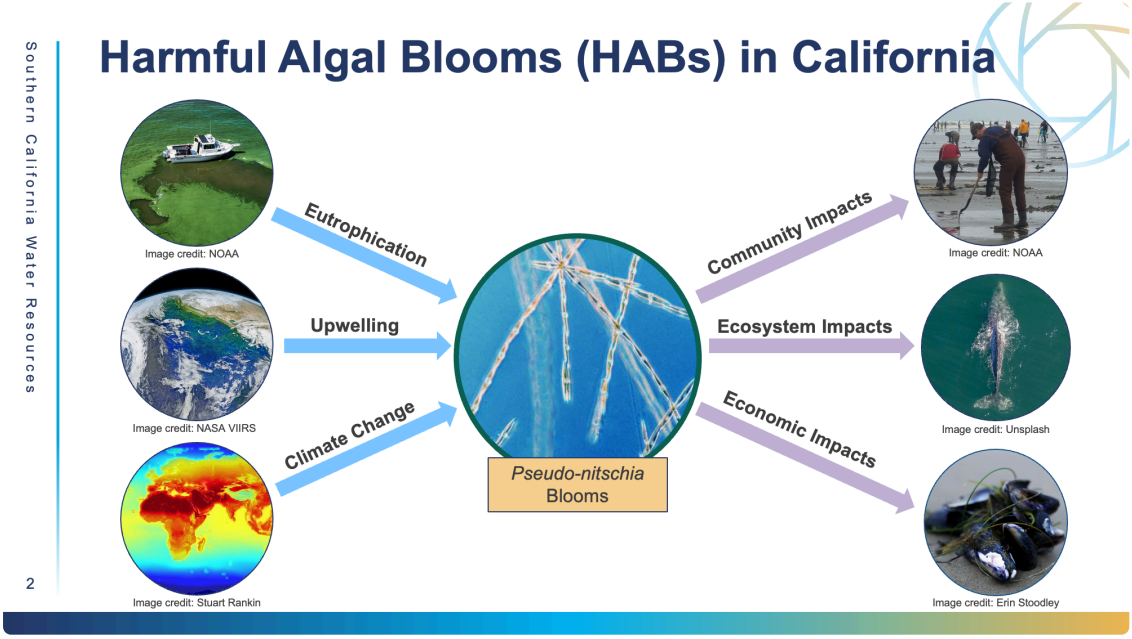
Harmful Algal Blooms

Chloe Belinsky, Karuna Chandran, Rachel Jiang, Lauren Novorska, Charlotte Rhoads

Project Overview

The novel SIT-FUSE framework has demonstrated immense potential when it comes to accurately predicting Harmful Algal Blooms. As a part of the NASA Earthrise Developer's Academy, the Southern California Water Resources team conducted a thorough evaluation of SIT-FUSE's potential. This project partnered with the Southern California Coastal Ocean Observing System and the California HAB Monitoring & Alert Program to integrate NASA Earth Observation satellite and sensor data for Harmful Algal Bloom (HAB) detection along the Southern California coast. We generated SIT-FUSE outputs from March 2024 to June 2025.

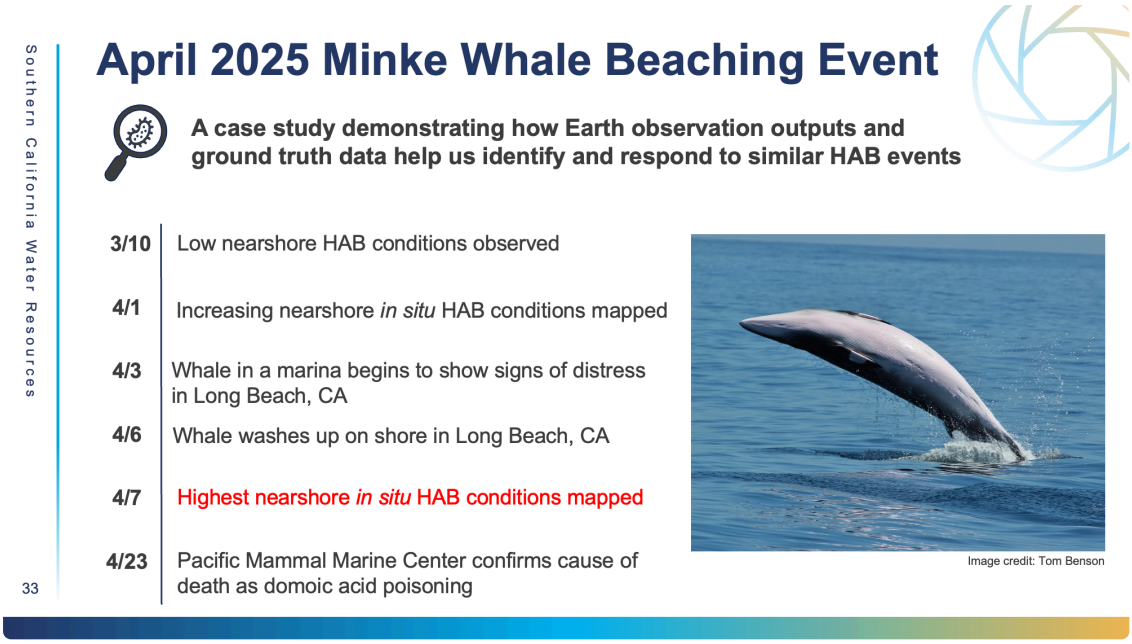
Why HABs?



Harmful algal bloom events (HABs) occur when algae rapidly proliferate and over accumulate. They can then pose a growing threat to community and ecosystem health and economic prosperity. These blooms are becoming more widespread due to nutrient enrichment from fertilizer runoff and seasonal upwelling, increased freshwater inputs, and climate-driven warming and ocean stratification (Carias et al., 2024; Trainer et al., 2020; U.S. National Office for Harmful Algal Blooms, n.d.) Pseudo-nitzschia spp., a toxic diatom, is of particular concern during HAB events off the coast of California, where blooms cause domoic acid accumulation in shellfish. Biomagnification of this toxin can result in Amnesic Shellfish Poisoning in humans and threaten California's aquaculture, fishing, and tourism industries, as well as the health of neighboring communities (Anderson et al., 2021).

In 2015, scientists recorded an unprecedented Pseudo-nitzschia spp. bloom on the U.S. West Coast. The Dungeness crab fishery, the most profitable fishery in the region, reported a $97.5 million decrease in revenue attributable to the blooms (Moore et al., 2020). Several other fisheries— including razor clam, rock crab, anchovy, and mussel— were also impacted, and elevated levels of marine mammal strandings and mortalities were reported (Bowers et al., 2018; Moore et al., 2020;). With 26.8 million people in coastal communities, 40 marine mammal species, 21 shorebird species, and 571 fish species, and extensive fishing and tourism industries, the health of California's coastal waters are vital (Harris et al. 2026; National Oceanic and Atmospheric Administration, n. d.)

The Minke Whale Beaching Case Study



On October 3, a minke whale began to show signs of distress in Long Beach, CA. On October 6, the whale ended up washing up on shore, and a few weeks later, the cause of death was determined to be: domoic acid poisoning. The whale incident was documented by a variety of news outlets and gained a lot of traction.

Connecting this incident back to SIT-FUSE, how can we use Earth Observation data to be able to remotely detect and predict HABs, so we may respond to similar incidents in a timely and effective manner?

What do SIT-FUSE outputs look like when visualized?

SIT-FUSE outputs, when visualized in a geospatial software (in this case ArcGIS), are an array of pixels stored within a GeoTIFF, each encoded with a specific value (0 to 5) based on how severe the framework predicts the bloom to be at that specific pixel.

For Phytoplankton conentration, the classes are as follows:

0 - Null (0-1000 cells/L)

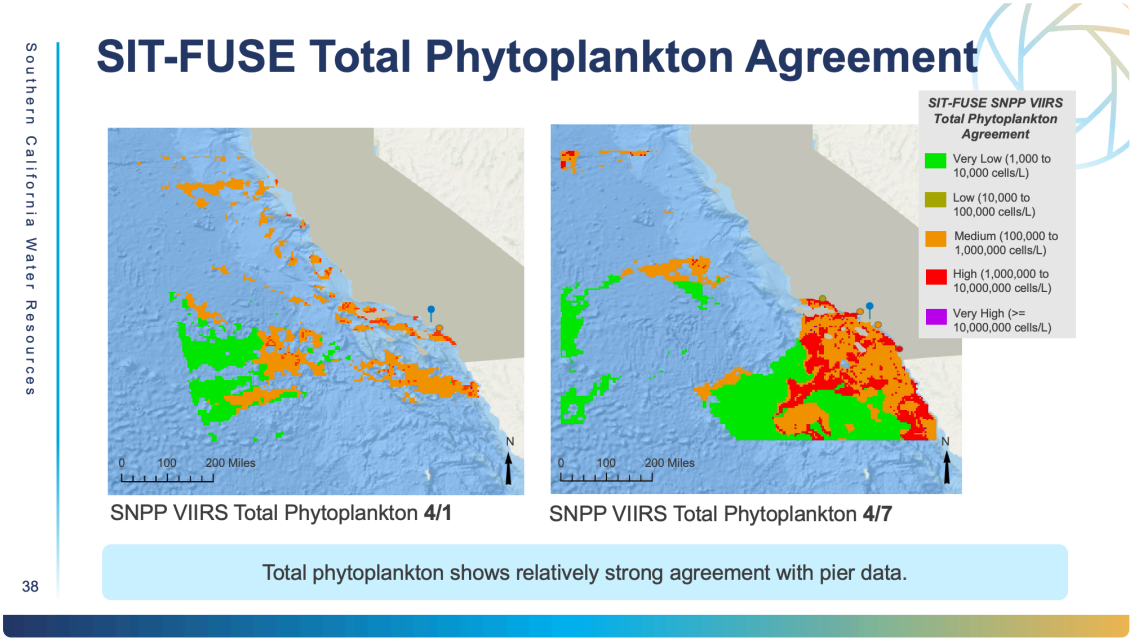
1 - Very Low (1,000 to 10,000 cells/L)

2 - Low (10,000 to 100,000 cells/L)

3 - Medium (100,000 to 1,000,000 cells/L)

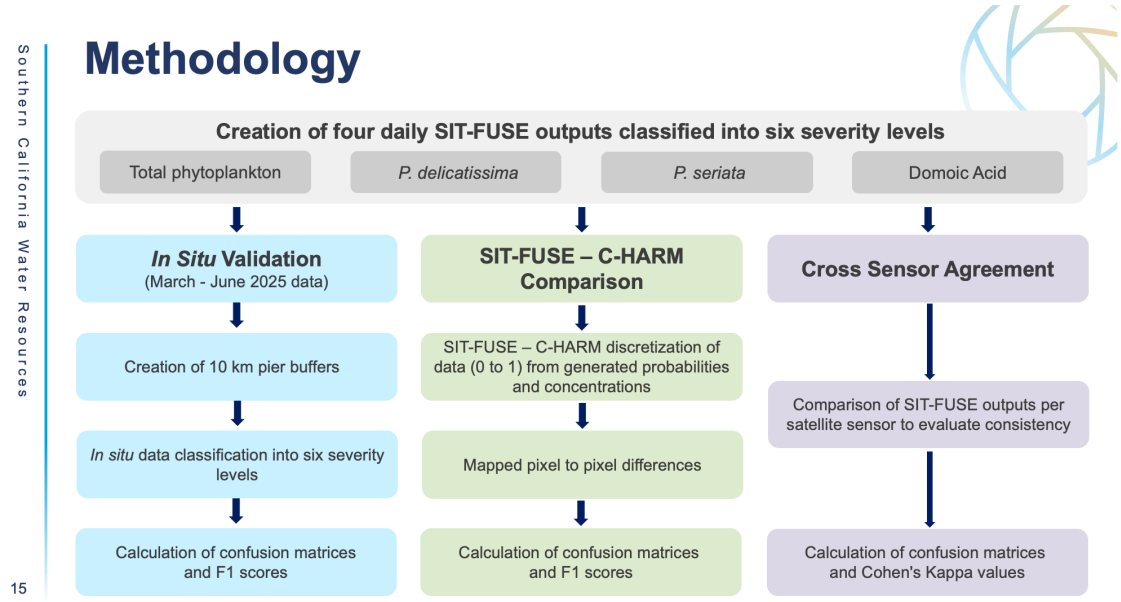
4 - High (1,000,000 to 10,000,000 cells/L)

5 - Very High (≥ 10,000,000 cells/L)



Thinking about the dates surrounding the Minke Whale Case Study, we visualized the SIT-FUSE outputs for April 1st and April 7th, where there was sufficient data from the Newport Pier, the closest pier to the whale beaching. We classified the different severity values into different colors, aligned with La Haye et al. 2026. We also overlaid *in situ* data from the piers, color coded the data, and created "buffers" around the piers with a 10 kilometer radius. Often, the SIT-FUSE model has difficulty predicting values close to the coast, so the buffer is to to ensure adequate overlap between the pier data and the SIT-FUSE outputs for validation.

Our Data Flow

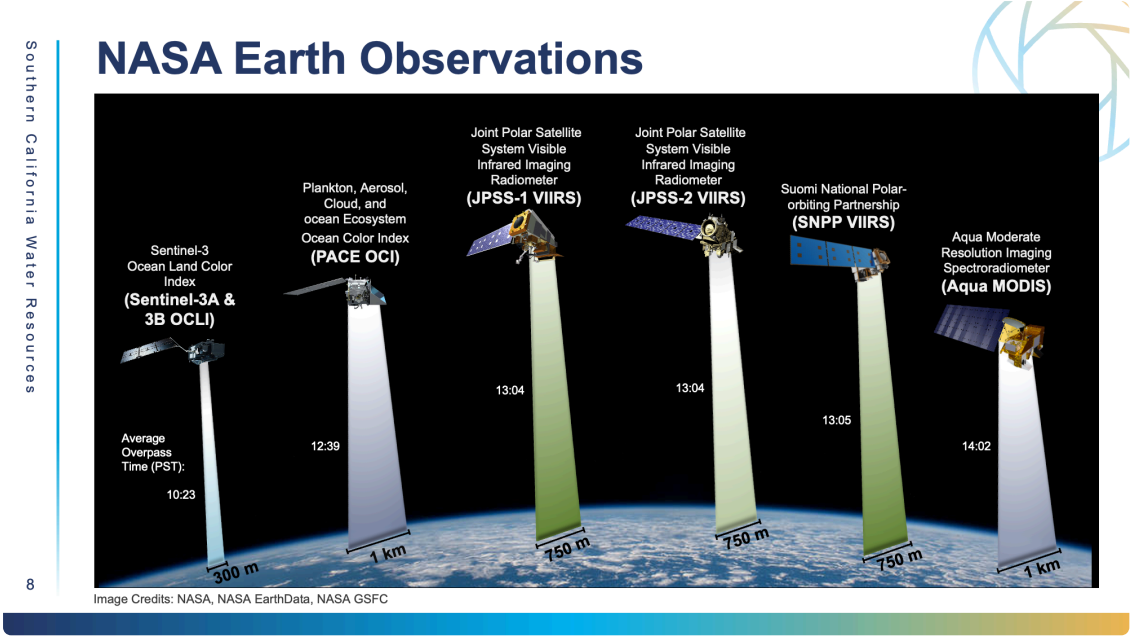


Our analysis was conducted on **three levels**:

1. A **validation** of the SIT-FUSE outputs with in situ pier data provided by Cal HABMAP. The goal is to assess how well the SIT-FUSE framework performs by comparing it to ground truth data.
2. A **comparison** of SIT-FUSE with the current California Harmful Algae Risk Model (C-HARM) model used for HAB detection to understand how well the two agree.
3. A cross-sensor **agreement** of all legacy sensors with one another to see which sensors perform similarly/differently for different data products.

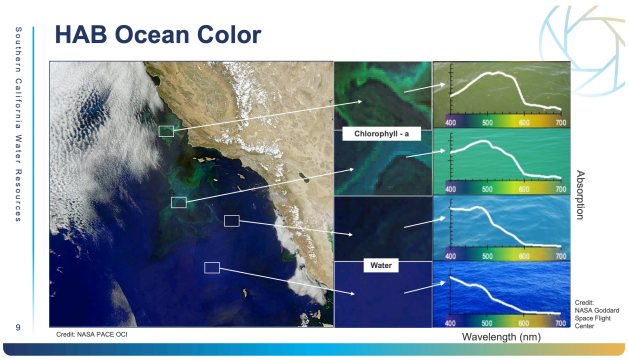
From these, we generated **four different products**:

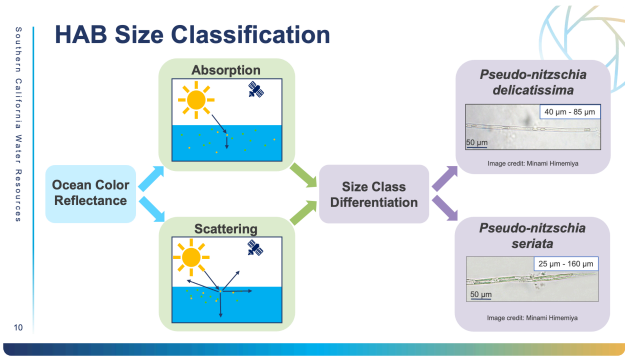
1. **Total Phytoplankton**, which has historically been the highest accuracy HAB detection product.
2. Two separate size classes of phytoplankton genus *Pseudo nitzschia* (P. spp): **P. seriata** and **P. delicatissima** (cells/L). The larger P. seriata has been historically shown to be more toxic than its smaller counterpart.
3. **Particulate domoic acid** (pDA), an entirely novel product (ng/mL). pDA is the most direct product in terms of being able to use outputs to accurately predict and quantify the toxicity of a bloom. This makes it a highly valuable product for stakeholders.



We analyzed outputs from **seven different legacy sensors**: PACE OCI, Sentinel-3A OLCI, Sentinel-3B OLCI, JPSS1 VIIRS, JPSS2 VIIRS, SNPP VIIRS, and finally Aqua MODIS

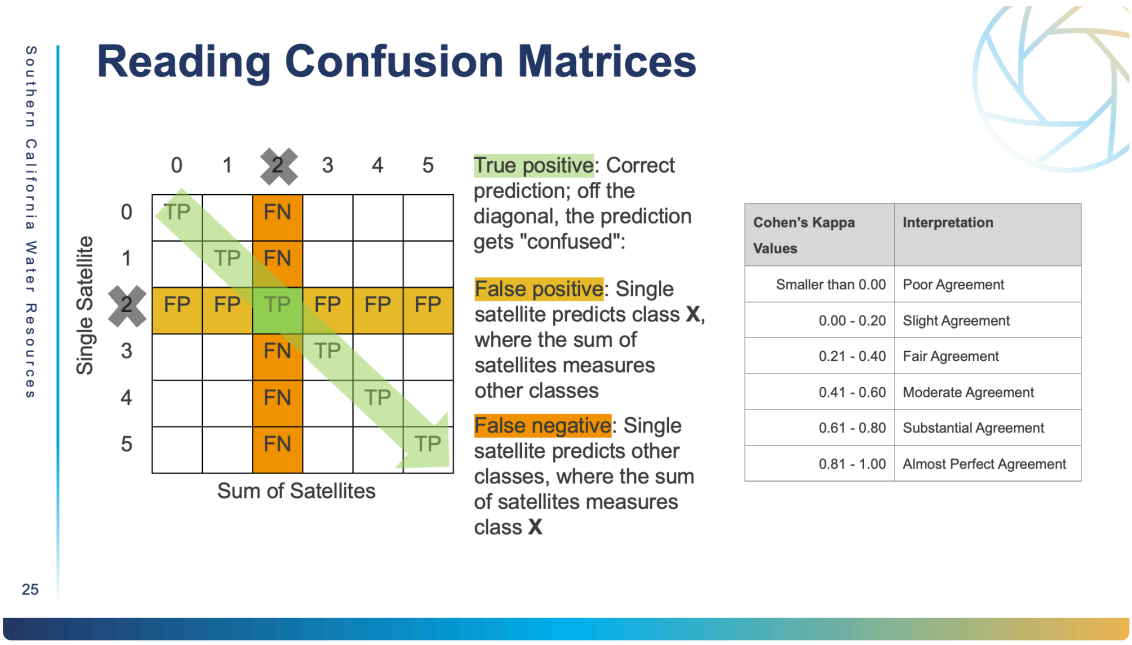
1. The Ocean Color Instrument **(OCI)** deployed on the Plankton, Aerosol, Cloud, ocean Ecosystem **(PACE)** platform
2. The Ocean and Land Color Instruments **(OLCI)** deployed on the **Sentinel-3A** and **Sentinel-3B** platforms
3. Visible Infrared Imaging Radiometer Suite **(VIIRS)** instruments deployed on the Suomi National Polar-orbiting Partnership **(SNPP)**, Joint Polar Satellite System-1 **(JPSS-1)** and Joint Polar Satellite System-2 **(JPSS-2)**
4. The Moderate Resolution Imaging Spectroradiometer **(MODIS)** instrument deployed on the **Aqua** platform.





NASA’s Earth Observing satellites provide near daily measurements of ocean color along the California coast. Because dense phytoplankton blooms change how light is absorbed and scattered in the water, Pseudo-nitzschia blooms can be detected from space, helping fill the spatial and temporal gaps left by *in situ* sampling.

1. Ocean color sensors measure water-leaving radiance to derive remote sensing reflectance, which essentially tells us how the ocean reflects light at different wavelengths. This reflectance is measured across multiple wavelength bands spanning the visible spectrum and analyzing the shape of this spectrum can shed light on the optical properties of what we're seeing in the water.
2. Traditionally, **chlorophyll-a** is used as a proxy to predict phytoplankton concentration. Chlorophyll-a absorbs blue and red while reflecting green, which is why water with high chlorophyll concentrations often appears green. This distinct pattern of absorption and reflectance creates a spectral 'fingerprint' that helps identify HAB indicators using satellite imagery.
3. Alongside multispectral remote sensing reflectance from **MODIS**, **VIIRS**, and **OLCI**, SIT-FUSE ingested hyperspectral reflectance from **PACE's Ocean Color Instrumen**t. Hyperspectral reflectance can is used to resolve the subtler spectral differences between size classes with a far wider variety of spectral bands and improved resolution.
4. Measurements from sensors like **PACE OCI** help distinguish different size classes of *Pseudo-nitzschia* through these optical signatures. This allows us to separate *Pseudo-nitzschia* from other materials in the water and to tell apart the larger ***Pseudo-nitzschia seriata cells*** from the smaller ***Pseudo-nitzschia delicatissima*** cells. The larger seriata group produces more domoic acid, so identifying size classes helps us understand the severity and potential toxicity of a bloom.



Cohen's Kappa Values	Interpretation
Smaller than 0.00	Poor Agreement
0.00 - 0.20	Slight Agreement
0.21 - 0.40	Fair Agreement
0.41 - 0.60	Moderate Agreement
0.61 - 0.80	Substantial Agreement
0.81 - 1.00	Almost Perfect Agreement

We used three primary modes of statistical analysis for this study: confusion matrices, F1 scores, and Cohen's kappa scores.

1. We constructed **confusion matrices** to quantify where different metrics predicted the same harmful algal bloom severity levels, providing counts of true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN) across all severity levels.
    a. Confusion matrices allow us to assess SIT-FUSE performance between sensors and against in situ data, including identification of systematic biases (such as overprediction or underprediction) within specific bins, across satellites, and generated products.
    b. True positives are along the diagonal, where both the *in situ* data and sensors successfully predicted the same value.
    c. For C-HARM comparisons, we generated confusion matrices and corresponding metrics on a scene-by-scene basis and aggregated these matrices temporally to assess consistency over the study period.
2. We then computed **F1 scores** from the constructed confusion matrices. We then computed precision, recall, and F1 scores from the constructed confusion matrices. Precision measures the accuracy of predictions; recall measures the framework's ability to predict all real conditions, and F1 scores are the harmonic mean of the two metrics. Finally, we calculated weighted F1 scores to account for the fact that different sensors had different numbers of "same day matchups". Same day matchups are when there was data coverage on a specific day of both sources of data being compared to one another.

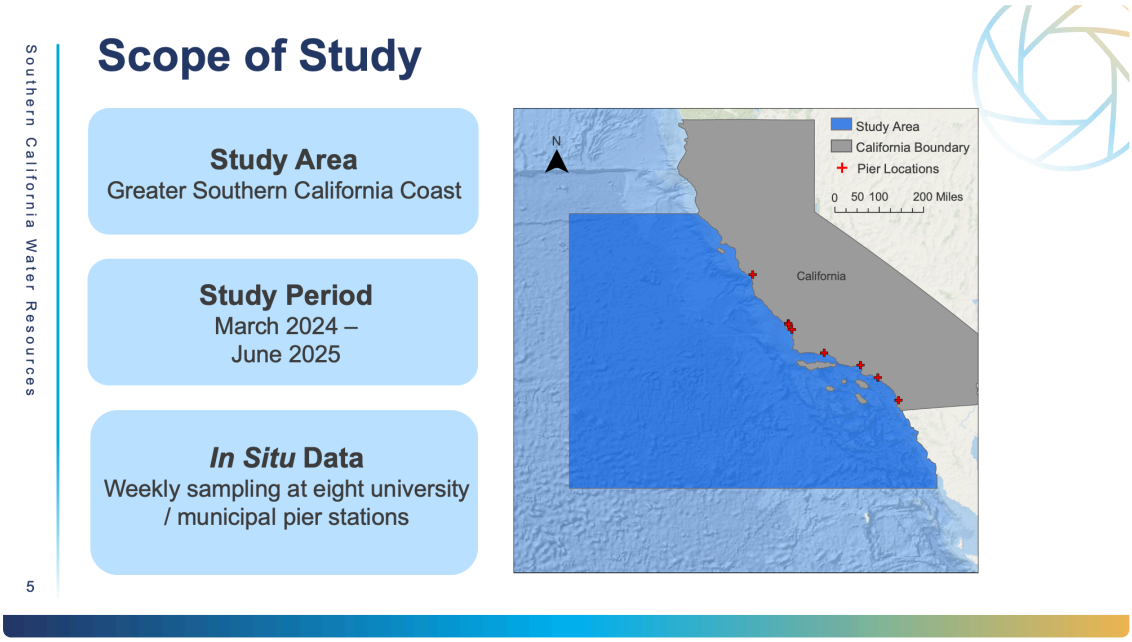
a. $precision = TP/TP + FP$
b. $recall = TP/TP + FN$
c. $F1 = 2\ x\ (precision\ x\ recall)\ /(precision + recall)$

3. We calculated quadratically weighted **Cohen's Kappa** coefficients (k) to evaluate consistency across sensors beyond chance and assigned penalties based on disagreement severity according to their distance apart. A Kappa score of under 0 indicates the comparisons were worse than chance and a score of 1 indicates perfect agreement between the sensors. We chose this metric as it accounts for the ordinal relationship between classes (ie. we can see how off one model is from another in terms of the class of HAB severity it predicts).

In Situ Validation

In Situ Data

Data Acquisition



1. We obtained *in situ* data through the California Harmful Algal Bloom Monitoring and Alert Program via the program website (https://calhabmap.org/↗). We selected pier data from eight locations along the Southern California coast and downloaded data across 2024-2025.
2. Parameters of interest include: cell counts (cells/L) of total phytoplankton, Pseudo-nitzschia (cells/L) broken down by size classes P. delicatissima and P. seriata, and particulate domoic acid (pDA, nanograms/L).

Data processing

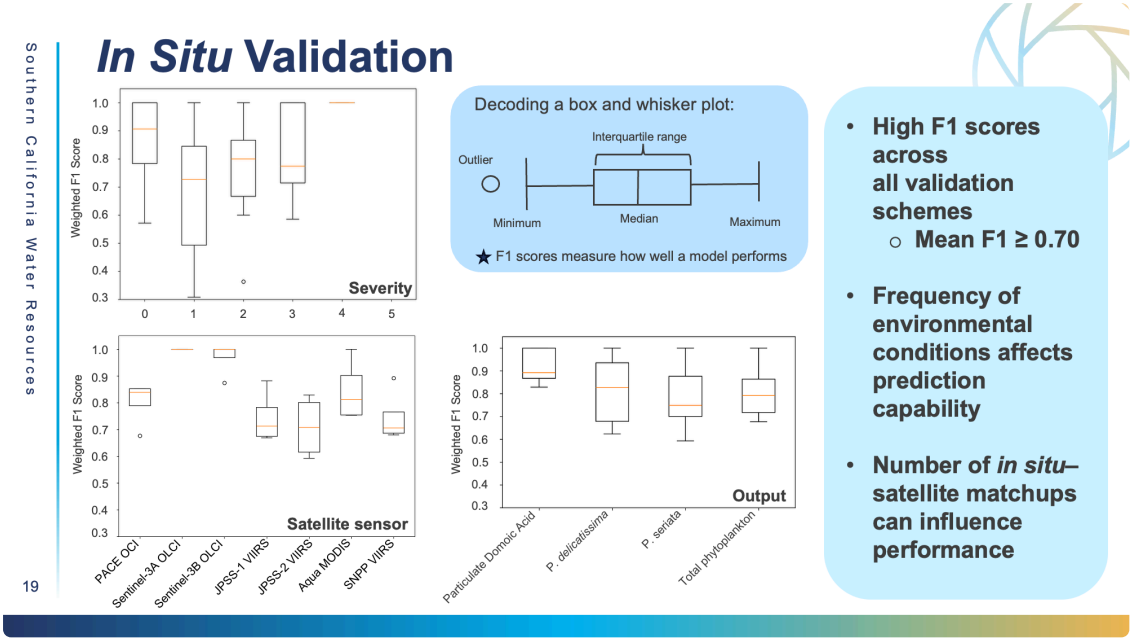
1. We processed Level 3 sensor inputs using the SIT-FUSE framework described by LaHaye, et al. (2026) for self-supervised representational learning and deep clustering. Level 3 satellite data is data that has been pre-processed and mapped to uniform space-time grid scales, making it easier to use. The lower the layer number, the less processed the data is.

2. SIT-FUSE is trained using self-supervised learning, meaning the data is organized into clusters by similarity that initially lack specific HAB severity labels. This is known as "context-free segmented data". In order to add that environmental HAB context to the data, we used the *in situ* database from March 2024 to February 2025 for context assignment to match the clusters with specific severity levels.
3. We performed context assignment on the coarser Layer 1 context-free segmented data, applied it to the finer-scale Layer 2 data, resulting in daily products (total phytoplankton, P. delicatissima, P. seriata, and particulate domoic acid) for all satellite scenes.
4. Due to the pier station's proximity to land, we defined a radius of 10 km (0.09°) as the threshold within which a same-day pixel–in *situ* matchup is considered valid. In other words, if a pixel overlaps with the radius around the pier, that particular matchup can be used for validation. We binned cell counts for total phytoplankton and size classes using a standardized categorical framework (LaHaye et al., 2026).
5. The SIT-FUSE framework has of six bins representing cell abundances ranging from not present to very high (≤10,000,000 cells/L).
6. We categorized particulate domoic acid concentrations similarly using a six-tier framework corresponding to ranges measured by Perry et al. (2023), ranging from not present to very high (≤1,000 ng/L). Across parameters, we referred to these categorical bins as severity levels in this study (0-5).

Data Analysis

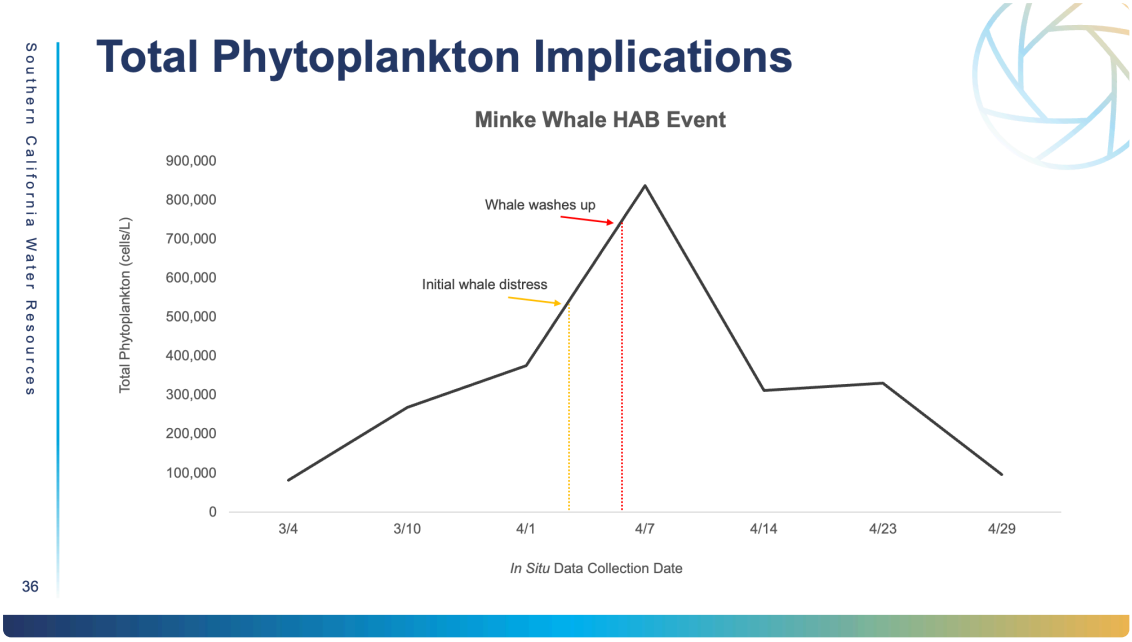
1. To assess SIT-FUSE performance, we constrained the generated outputs and *in situ* observations to our study period from March 2025 through June 2025 and evaluated them in a matchup framework. We kept the 10 km (0.09°) radius around the pier station as the spatial threshold for valid pixel–*in situ* comparisons, which matches with the above criteria used in context assignment.
2. For each satellite scene within the validation period, we extracted the SIT-FUSE–predicted severity level at the pier station location and compared it against the same day *in situ* observation. We conducted comparisons for every product individually, with agreements evaluated across all defined severity levels. We summarized the resulting matchups using confusion matrices and F1 scores.

Data Results



1. We compiled the F1 scores grouped by sensor platform, SIT-FUSE product, and severity class into box and whisker plots. The F1 Scores were high across the board, with the mean F1 score >/= 0.70.
2. **Sentinel-3A and -3B** performed the best by instrument, when comparing the medians, while JPSS1 and JPSS2 VIIRS were the weakest.
3. Out of all the products, **pDA** performed the best by far and had the highest F1 score, with an F1 score around 0.9, while the phytoplankton metrics remain relatively comparable.
4. A **severity level of 4** had the highest F1 score median, compared to a **severity level of 1**, with a large range and a low median of data.
5. The frequency of environmental conditions, as well as the number of *in situ* satellite matchups can have a large impact on the performance of the model.

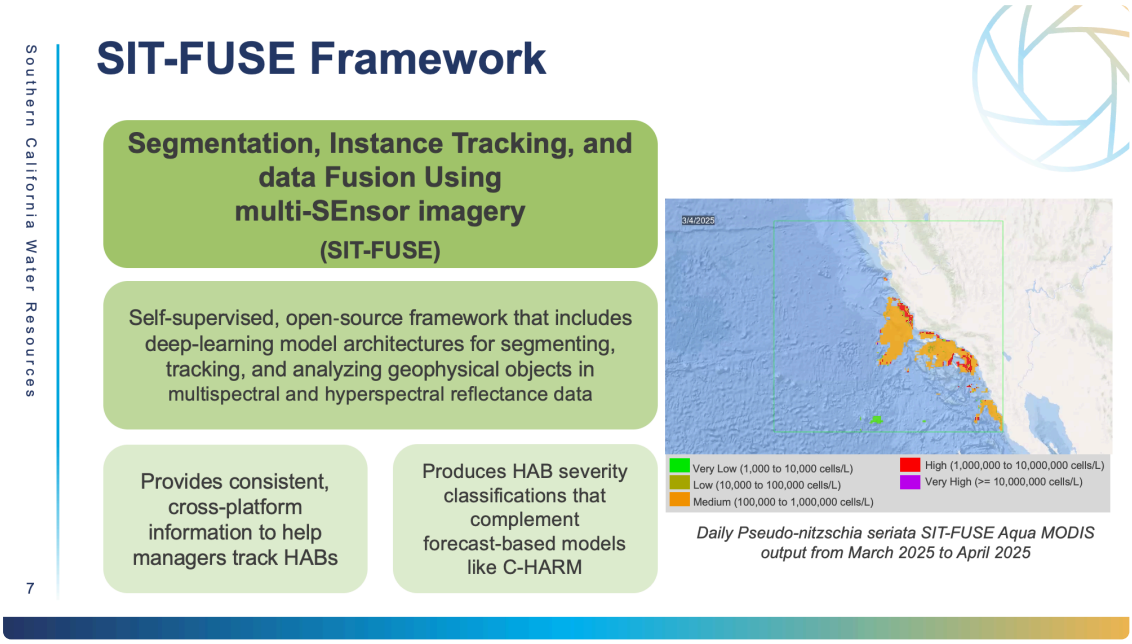
Case Study Implementation



1. To visualize the Minke Whale case study, we chose to use **SNPP VIIRS** as our primary sensor. SNPP VIIRS has a moderate overall F1 score and an good amount of *in situ* data by the Newport Pier, the closest pier to the site, which was beneficial for satellite-*in situ* matchups. SNPP VIIRS also showed high predictive total phytoplankton outputs in ArcGIS.
2. When conducting a thorough validation of SIT-FUSE outputs, considering a) the abundance of data, b) predictive capabilities of each individual sensor, and c) the accuracy of different HAB severity labels is necessary for a thorough evaluation.

Validation Code Introduction

Where to access the *in situ* validation code for our project?



Daily Pseudo-nitzschia seriata SIT-FUSE Aqua MODIS output from March 2025 to April 2025

1. Start from https://github.com/nlahaye/SIT_FUSE ↗ main page
2. Inside this branch, go to src/sit_fuse > pipelines > habs > hab_validation_analysis_ERDA.py. This is where SIT-FUSE validation analysis metrics (F1 Scores and box plots) are created
3. If one has the output file, this code can simply be run by using: python src/sit_fuse/pipelines/habs/hab_validation_analysis_ERDA.py "D:\Users\charlotte.rhoads\Downloads\output_vals.pkl"
    a. The pathway should be modified based on the individual's file pathway where the output_vals.pkl file is stored.
4. Within the habs folder, hab_post_inference_pipeline.py is what creates the output files that are then analyzed in the validation analysis tab.

How does the output code work?

Once the framework has completed the "inference" stage using the satellite imagery, the script takes the HAB severity classes assigned to each pixel and attaches all the useful environmental metadata. This data is organized and structured into a Python dictionary which is then saved as a file called output_vals.pkl. Pickle (pkl) files are useful when you want to avoid rerunning outputs that have already been generated and save them to a file that it can be simply loaded into memory using the validation code.

How does the validation code work?

The validation code then takes this output file. This stores many validation records which are each single comparisons between between model/*in situ* matchups which need to be converted to 1s and 0s depending on predicted HAB or no HAB labels. The script goes through each record one by one, determining agreement categorized as true positive, true negative, false positive, false negative. The validation script is able to process each sample, compare and record agreement with *in situ* data, and summarize this information for the whole dataset. From there, it is able to generate confusion matrices, F1 Scores and Cohen's Kappa values.

C-HARM Comparison

C-HARM Comparison

Data Acquisition

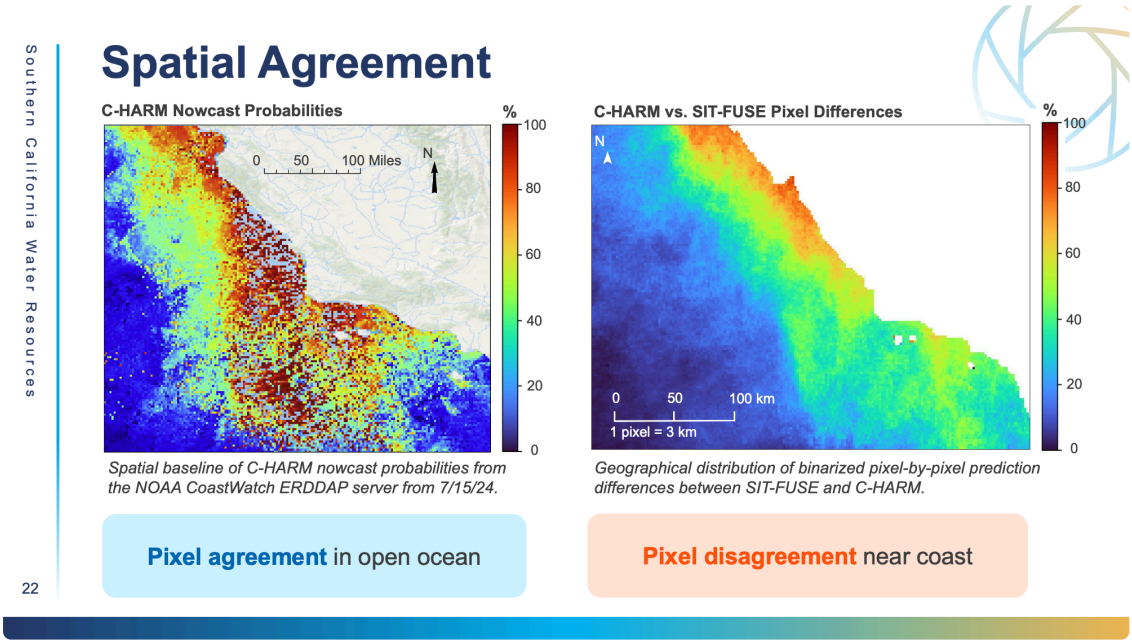
1. We acquired C-HARM probabilities for Pseudo-nitzschia and domoic acid from the National Oceanic and Atmospheric Administration’s (NOAA) Environmental Research Division's Data Access Program (ERDDAP) portal.
2. C-HARM generates daily nowcasts (predicted daily conditions) and 3-day forecasts for the probability that the concentration of Pseudo-nitzschia blooms exceeds 10,000 cells/L, pDA level (probability of DA cell abundance ≥ 500 ng/L in phytoplankton), and cellular toxicity (probability of DA ≥ 10 pg/cell of P. spp) (LaHaye et al., 2026).

Data Processing

1. In order to be able to compare SIT-FUSE’s self-supervised HAB detections with C-HARM’s nowcast probability predictions, we adopted the grid matchup and statistical validation workflow established by LaHaye et al. (2026) for our study scope. Daily continuous C-HARM probabilities (0.0 to 1.0) were aligned with their SIT-FUSE satellite counterparts, on days where there was a satellite overpass matchup. .
2. Both datasets were then split into binary layers representing active Bloom (1) and No Bloom (0) based on specific predetermined thresholds. For C-HARM’s Pseudo-nitzschia probabilities, the threshold categorized anything with a probability of 0.75 or over of having greater than 10,000 cells/L of P spp. as a Bloom (1). Similarly, a ≥0.75 threshold was applied to C-HARM's particulate domoic acid probabilities, to denote Bloom (1) events as ≥500 ng/L in phytoplankton. For SIT-FUSE, all values ≥2 were categorized as a Bloom (1), since 10,000 cells/L was the lower bound for this class's bin (LaHaye et al., 2026).
3. Coordinates pointing to invalid pixels due to landmasses, non-overlapping pixels or heavy cloud cover for example, were assigned an identifier of -1.0 to represent the lack of data at that pixel and masked out to isolate a strictly shared observational area consistent across both C-HARM and SIT-FUSE.

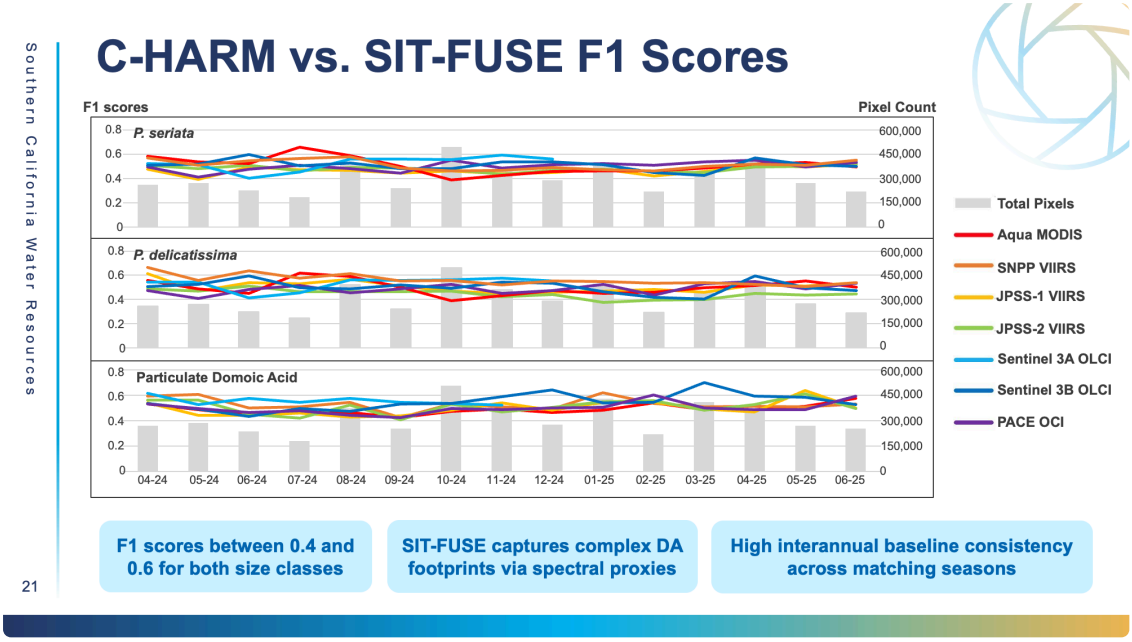
4. Because predictive models like C-HARM and satellite observations ingested by SIT-FUSE have fundamentally distinct spatial resolutions, we performed a spatial resampling to achieve pixel matchups on a pixel-by-pixel basis. C-HARM outputs data on a regional ocean model grid of ~3 km resolution, while SIT-FUSE's products are at a standardized resolution of 4 km.
5. This interpolation utilized the nearest-neighbor k-dimensional tree resampling algorithm, which quickly is able to assign values to a new grid cell based on the closest neighboring cell. This algorithm looks at points within a 10 km radius to ensure the highest possible spatial coverage across the resampled area without creating gaps in data near grid intersections. The resulting spatial array isolates the localized geographical distribution of prediction differences to focus on the Southern California coastline study area.

Data Analysis



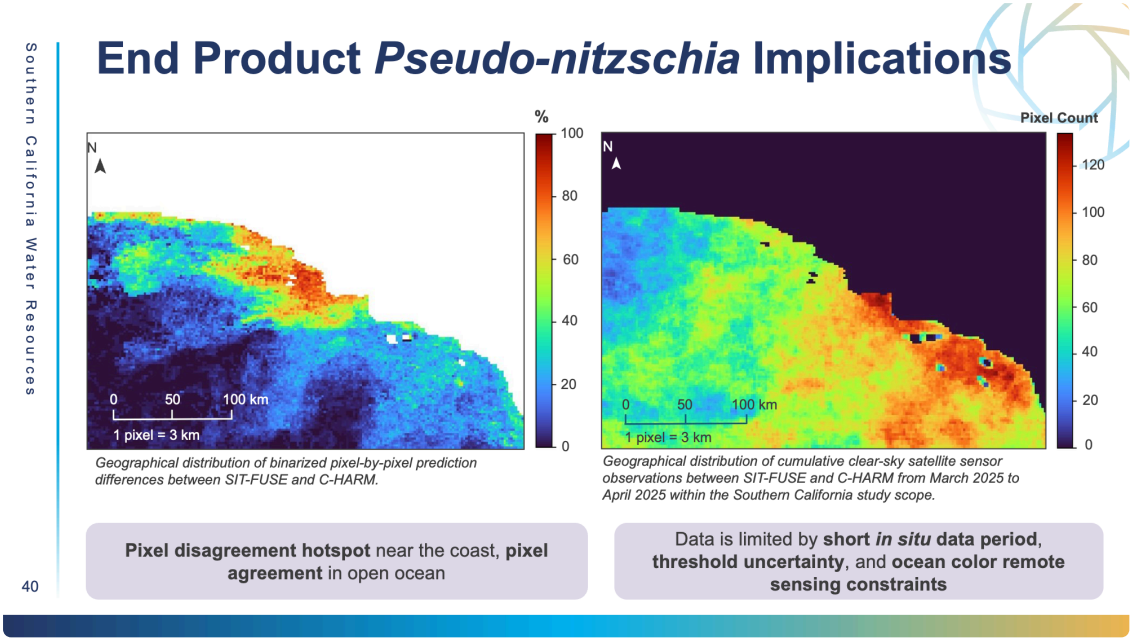
1. We compared the binarized SIT-FUSE and C-HARM outputs on a daily, pixel-by-pixel basis. From the pixels considered valid, we generated cumulative 2×2 binary confusion matrices to determine TN, FP, FN, and TP values, alongside daily F1-scores.
2. We then compiled these cross-sensor metrics into a standardized summary database to evaluate long-term model agreement.

Data Results



1. The figure above illustrates the quantitative validation between C-HARM and SIT-FUSE over a 14-month period using F1 scores. For both Pseudo-nitzschia size classes in the top panels, scores are consistent in the 0.4 to 0.6 range.
2. Given that PACE utilizes hyperspectral imagery rather than multispectral data, it is exciting that SIT-FUSE's known PACE capabilities have expanded to achieve clear consistency against C-HARM as well as older sensors throughout this new study period.
3. In the bottom panel for particulate domoic acid, scores look almost identical to the scores tracking the physical algae cells themselves. While a satellite cannot physically determine chemical toxins from orbit, the DA consistency with size classes proves SIT-FUSE successfully extracts the underlying environmental signatures associated with high-toxicity events.
4. Looking at the gray background bars, we included our total pixel count to show the sheer volume of valid data points ingested to produce these scores per month. Comparing June 2024 to June 2025, our data volume and F1 scores return to the exact same baseline. This year-over-year consistency shows that SIT-FUSE has truly learned the cyclical patterns of the Southern California upwelling ecosystem.
5. C-HARM overall had the highest agreement with SIT-FUSE closer to a “belt” along the coastline, with values around 0.8 and farther from the main ocean, with values around as low as 0.

Case Study Implementation

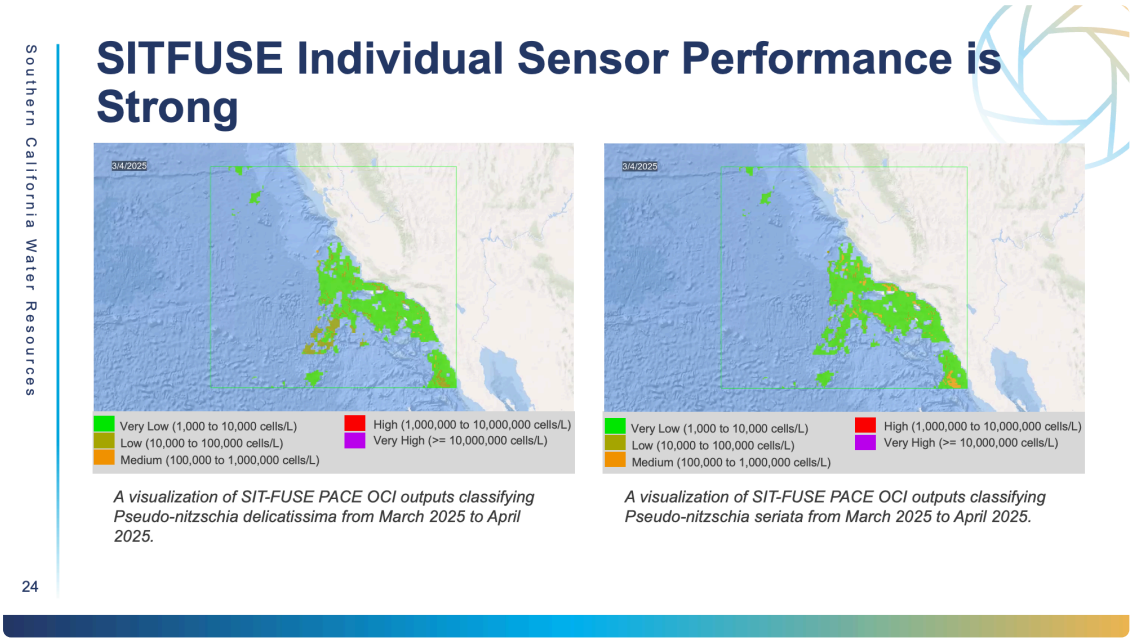


1. Looking at the case study time period from March to April 2025, with C-HARM, there tends to be a pixel disagreement hotspot **near the coast**, and pixel agreement in the **open ocean**.
2. Data is often limited by short ***in situ*** **data** period, **threshold** uncertainty, and **ocean color** remote sensing constraints.

Cross Sensor Agreement

Cross Sensor Agreement

Data Processing (Creation of Time Series Visualization)

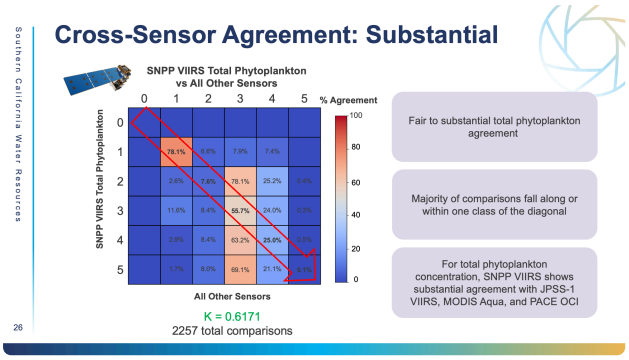


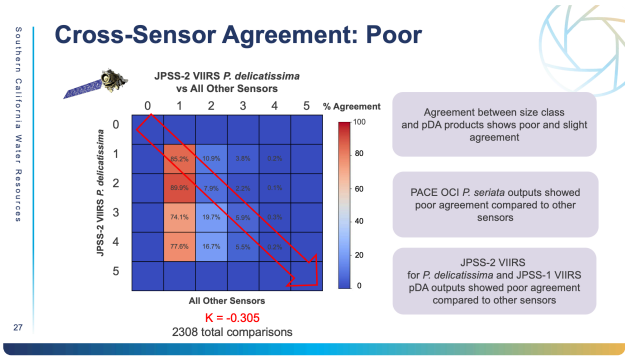
1. To visualize the SIT-FUSE products, we first evaluated the model outputs and bloom progression for Pseudo-nitzschia delicatissima, P. seriata, total phytoplankton, and domoic acid to create a per-sensor continuous time series of the outputs generated by the SIT-FUSE model. This would allow us to be able to visualize the bloom spread.
2. In ArcGIS Pro (Version 3.7.1), we added SIT-FUSE rasters to a mosaic dataset (a collection of SIT-FUSE outputs, each raster in the collection representing one specific date), and we rebuilt the Value Attribute Table for each of these rasters, looping these steps over all SIT-FUSE output rasters.
3. We then created a mosaic dataset for each sensor and each SIT-FUSE product and populated them with their corresponding SIT-FUSE output rasters. We set the resampling type to Nearest Neighbor to preserve the integrity of categorical values and set our time slider to match the temporal range of the data. When ArcGIS has to estimate new pixel values, Nearest Neighbor resampling has pixels copy values of the nearest neighboring pixels.
4. We matched our visualization style and severity class to the methodology outlined by LaHaye et al. (2026).

Data Analysis

1. We assessed agreement between sensors for each output product to determine how consistently SIT-FUSE classified the same location and date when it processed data from two different platforms.
2. For every unique pair of sensors, we identified sets of rasters sharing common dates between the two sensor folders and kept only dates with an output from both sensors for comparison. We masked pixels with a value of –1 whenever there was a lack of data, from both raster sets. The remaining valid pixels were flattened into vectors that were paired with one another and processed through Python library scikit-learn's “confusion_matrix” function using a label set of [0, 1, 2, 3, 4, 5]. This allowed for easy comparison between severity values of two different sensors.
3. We then generated a confusion matrix for each individual overlap date. We also summed the per-data matrices into a single total confusion matrix for each sensor pair and output product (pairwise), which allowed us to assess agreement both scene-by-scene and in total across the full comparison period.
4. In addition to the pairwise sensor comparison, we compared each sensor against the pooled outputs of all the other sensors for a given species. In other words, we compared sensors versus sensors, but also compared one sensor against all other sensors. For each target sensor, we identified overlapping against every other sensor, and we applied the same confusion matrix process described above to each date.
5. For both the sensor versus sensor and one-versus-all comparisons, we visualized the total confusion matrix as a heatmap, and normalized each row of the heatmap so that each row summed to 100%. This normalization shows that for a given sevreity class assigned by the reference sensor, the percentage of pixels each comparison sensor placed in every other class. Each cell reports both the raw pixel count and the row-normalized percentage.
6. We decided to measure the agreement using a quadratically weighted Cohen's kappa. We computed this once per sensor pair/per species for the pairwise comparison and once per target sensor/per species for the one-vs-all comparison. The purpose of quadratic weighting is to penalize disagreement between sensors in proportion to the squared distance between the two assigned classes. Because of this, one-class discrepancy (e.g., Low misclassified as Medium) contributes substantially less to the error than a multi-class discrepancy (e.g., Not Present misclassified as Very High).

Data Results

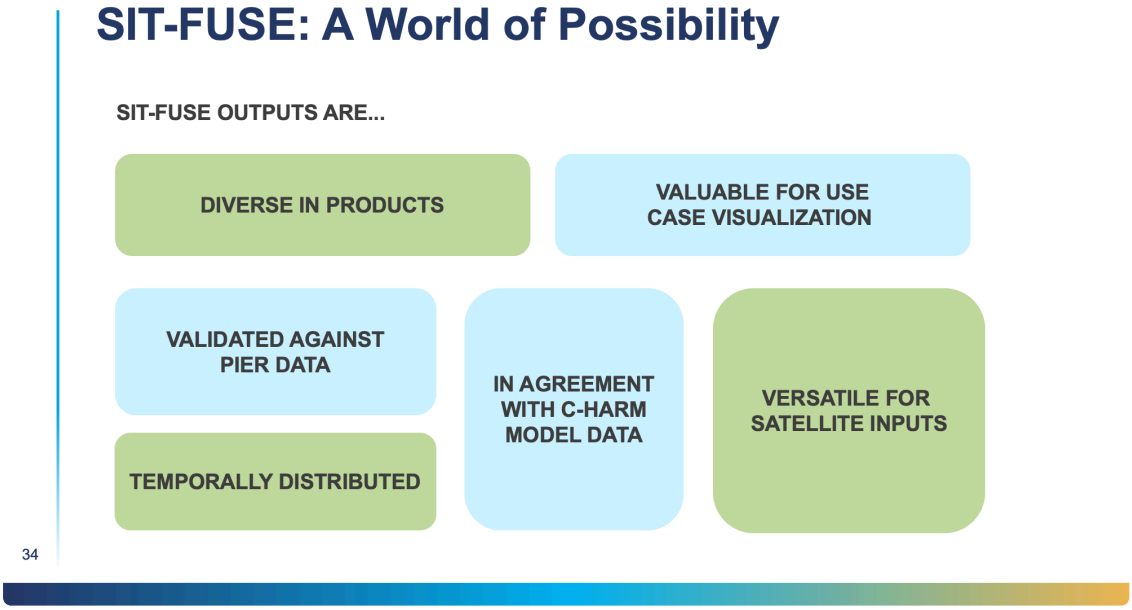




1. Cross sensor comparisons of total phytoplankton concentration showed moderate to substantial agreement across platforms overall, with Cohen's Kappa values ranging from k=0.2661 to k=0.7251. **SNPP VIIRS** reflected the strongest agreement across sensors (Fig 4.1).
2. When **total phytoplankton** concentration was compared against all other sensors (n=2,257), the Cohen's Kappa value was 0.6171, indicating substantial agreement.
3. The confusion matrix showed that the majority of comparisons fall along or within roughly one class of the diagonal, which indicates a true positive , with **class 1** showing 78.1% agreement and **class 3** showing 55.7% agreement.
4. Comparisons between SNPP VIIRS and individual individual sensors support this pattern as well, as agreement between SNPP VIIRS and JPSS1 VIIRS (k=0.7251) (Fig. 4.2), MODIS Aqua (k=0.6496) (Fig. 4.3), and PACE OCI (k=0.6493) (Fig. 4.4) reflect substantial agreement across platforms both carrying the same instrument and not. Overall, these results highlight how total phytoplankton concentrations are reasonably consistent across sensor platforms.
5. In contrast to agreement between total concentration products, agreement collapsed between size class and toxin classification products. Cohen's kappa values ranged between k=-0.0247 and k=0.1799 for **Pseudo nitzchia seriata**, k=-0.0757 and k=0.2296 for **Pseudo nitzchia delicatissima**, and k=-0.507 and k=0.1082 for **particulate Domoic Acid**.

6. In particular, **PACE OCI Pseudo nitzchia seriata** values compared against all other sensors (n=2,152) reflected a kappa value of k=0.0342 (Fig. 5a.1). **JPSS2 VIIRS delicatissima** classification compared against all other sensors (n=2,308) reflected a kappa value of -0.0305 (Fig. 5b.1) and **JPSS2 VIIRS particulate domoic acid** compared against all other sensors (n=2,333) reflected a kappa of -0.0351 (Fig 5c.1). Both values are below 0, indicating that agreement is marginally worse than what would be expected by random chance.
7. For all three, the majority of comparisons cluster into **single vertical classes**, with the majority of paired observations falling into **class 1** regardless of which individual sensor and class it was being compared to. This pattern held across all individual sensor pairings for seriata, delicatissima, and particulate domoic acid.

Case Study Implementation

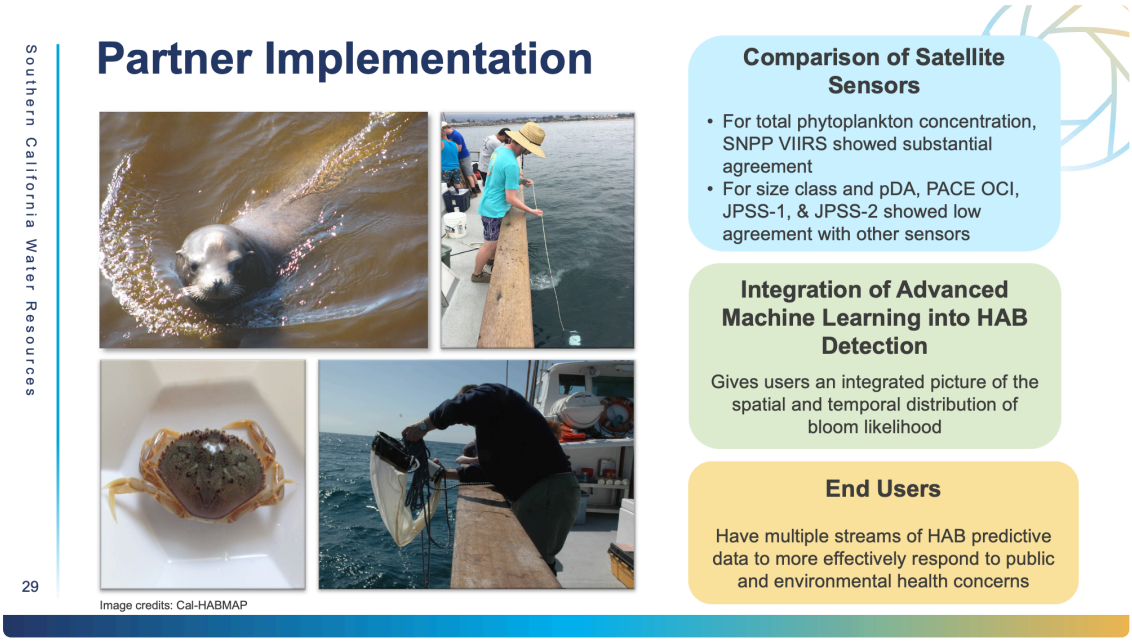


1. Creating SIT-FUSE output time series can allow stakeholders to visualize bloom spread and severity over time in a clear, understandable format. They can be applied to a variety of products, compared against pier buffered data, and used with any of the legacy sensors.
2. Understanding sensor agreement can help with future efforts to combine like sensor inputs into more integrative HAB predictive models in order to better respond to environmental and community health threats.

Conclusions

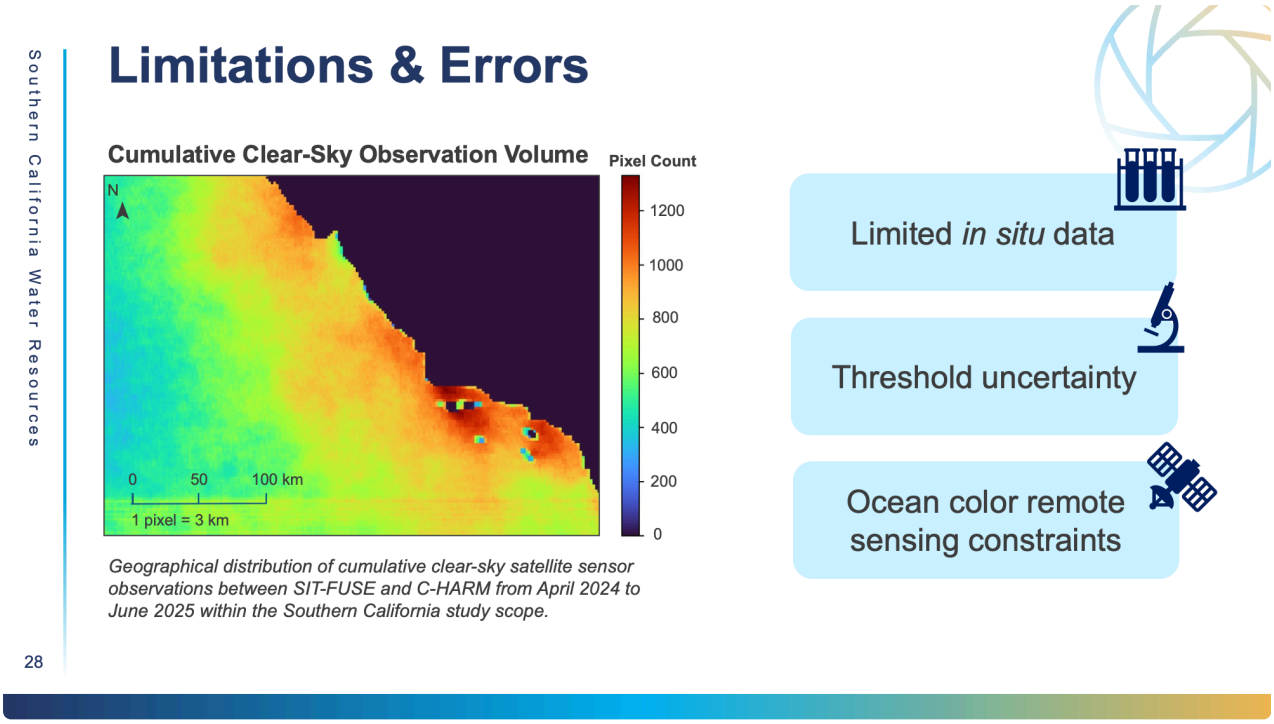
Key Takeaways

Partner Implementation



The primary goal of our project was to give end users a more integrated picture of the spatial and temporal distribution of bloom likelihood. Understanding which sensors have stronger agreement with one another can allow for new diverse, integrated models with better accuracy and spatial and temporal coverage. In the future, Earth observation-based frameworks like SIT-FUSE could effectively complement existing models such as C-HARM to provide end users with multi-stream predictive HAB data to effectively respond to public and environmental health concerns facing Southern California communities.

Limitations and Errors

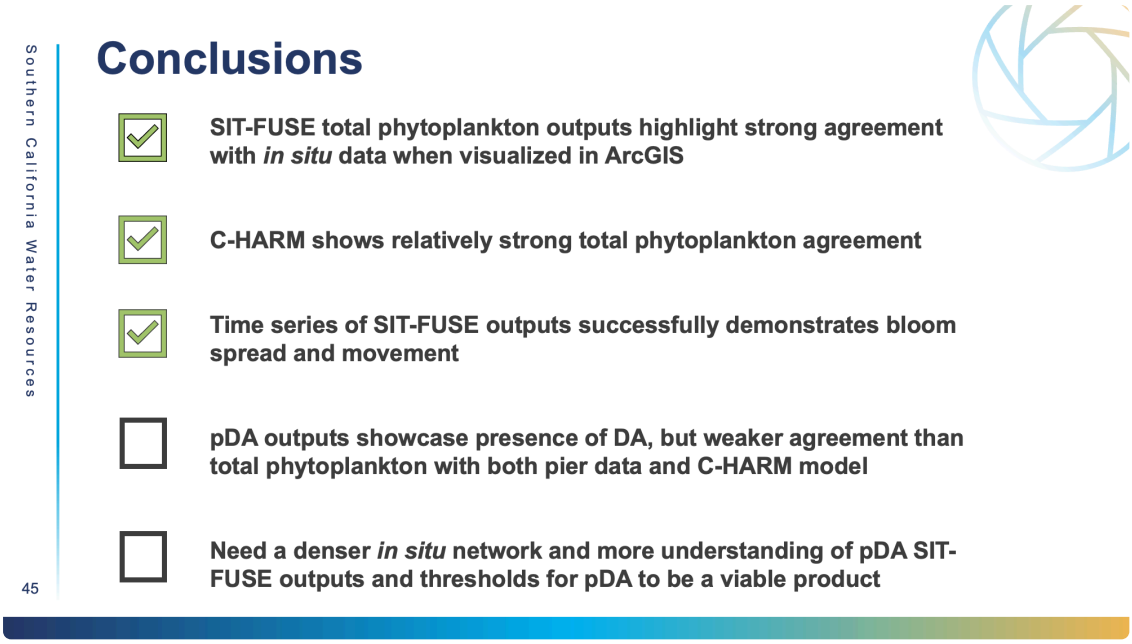


Geographical distribution of cumulative clear-sky satellite sensor observations between SIT-FUSE and C-HARM from April 2024 to June 2025 within the Southern California study scope.

There are a few sources of possible errors in our study:

1. We used a 10 km radius to define valid satellite–*in situ* matchup to accommodate the fixed pier observations nearshore. While this is a commonly used approach, it can introduce spatial mismatch. Additionally, we were limited to 8 pier stations that measured our parameters of interest during our study period.
2. Second, we applied a C-HARM probability threshold of 0.75 to indicate the presence of Pseudo-nitzschia, consistent with previous studies. However, this threshold is generalized and not directly tied to local bloom dynamics, highlighting the need for further refinement and validation.
3. Lastly, matchup availability between satellite and *in situ* data is limited. Differences in satellite overpass and *in situ* collection frequency mean that many satellite scenes lack corresponding *in situ* measurements; this results in relatively few same-day matchups for context assignment and validation. We also encountered environmental and observational limitations, preventing accurate retrieval of ocean color in optically complex waters.

Case Study Conclusions



1. Overall, SIT-FUSE **total phytoplankton** outputs highlight strong agreement with *in situ* data when visualized in ArcGIS. C-HARM also shows relatively strong total phytoplankton agreement. The time series of SIT-FUSE outputs successfully demonstrates bloom spread and movement.
2. In terms of the **pDA** products, pDA outputs showcase presence of DA, but weaker agreement than **total phytoplankton** with both the pier data and C-HARM model for this case study snapshot. We need a denser *in situ* network and more understanding of pDA SIT-FUSE outputs and thresholds for pDA to be a viable product

Final Thoughts

Overall, results support SIT-FUSE ingestion of Earth Observation data as a viable complimentary strategy for HAB monitoring in California to existing models such as C-HARM. The current C-HARM model utilizes data from VIIRS, but cross sensor comparison, which sensors have stronger agreement with one another can allow for new diverse, integrated models with better accuracy and spatial and temporal coverage.

1. **SIT-FUSE Validation: Sentinel-3 OLCI A/B OLCI** and **PACE OCI** show strongest performance against *in situ* validation, with F1 scores of 1 and 0.84, respectively. While all SIT-FUSE products performed well (F1 ≥0.75), **particulate domoic acid** (pDA) showed the strongest *in situ* validation (F1= 0.89) overall, which bodes well for further research of pDA as a future product.
2. **C-HARM Comparison:** Long-term coastal disagreement between C-HARM and SIT-FUSE concentrates in northern waters, indicating satellite retrieval errors in regions that are optically complex and hard to analyze. **Size classes** and **pDA** show high overall consistency with C-HARM across the seasons, proving SIT-FUSE captures cyclical seasonal upwelling.
3. **Cross Sensor Agreement: Total phytoplankton** concentration shows substantial agreement across sensors looking at Kappa Scores (up to k=0.73), supporting SIT-FUSE's capabilities across sensors to identify blooms. **Size classes** and **pDA** show poor cross-sensor agreement, suggesting this framework would benefit from combining like sensors.

SIT-FUSE opens a world of possibilties for future applications of Earth Observation data for HAB detection and many other environmental use cases.

Bibliography

Anderson, D. M., Fensin, E., Gobler, C. J., Hoeglund, A. E., Hubbard, K. A., Kulis, D. M., Landsberg, J. H., Lefebvre, K. A., Provoost, P. Richlen, M. L., Smith, J. L., Solow, A. R., & Trainer, V. L. (2021). Marine harmful algal blooms (HABs) in the United States: History, current status and future trends. Harmful Algae, 102, Article 101975. https://doi.org/10.1016/j.hal.2021.101975.

Bailess, A., Baetge, N., Barnard, A., Tufillaro, N., Behrenfeld, M., Bill, B., Kudela, R., Graff, J., & Kavanaugh, M. (2026). Hyperspectral remote sensing for harmful algal bloom detection: Pseudo-nitzschia in the Northeast Pacific. BioRxiv. https://doi.org/10.64898/2026.02.24.707776. ↗

Bowers, H. A., Ryan, J. P., Hayashi, K., Woods, A. L., Marin III, R., Smith, G. J., Hubbard, K. A., Doucette, G. J., Mikulski, C. M., Gellene, A. G., Zhang, Y., Kudela, R. M., Caron, D. A., Birch, J. M., & Scholin, C. A. (2018). Diversity and toxicity of Pseudo-nitzschia species in Monterey Bay: Perspectives from targeted and adaptive sampling. Harmful Algae, 79, (129–141). https://doi.org/10.1016/j.hal.2018.08.006.

California Department of Technology. (2024). CA state boundary [Data set]. California Open Data Portal. https://data.ca.gov/dataset/e212e397-1277-4df3-8c22-40721b095f33/resource/3db1e426-fb51-44f5-82d5-a54d7c6e188b/download/ca_state.zip.

Carias, J., Vásquez-Lavín, F., Barrientos, M., Ponce Oliva, R. D., & Gelcich, S. (2024). Economic valuation of Harmful Algal Blooms (HAB): Methodological challenges, policy implications, and an empirical application. Journal of Environmental Management, 365, Article 121566. https://doi.org/10.1016/j.jenvman.2024.121566.

Harris, J.L., Waters T., & Zulian M. (2026). California's 2026 Coast and Ocean Assessment. Ocean Science Trust. https://opc.ca.gov/wp-content/uploads/2026/03/2026-Coast-and-Ocean-Assessment.pdf.

LaHaye, N. J., Luis, K. M., & Gierach, M. M. (2026). Fusing multi- and hyperspectral satellite data for harmful algal bloom monitoring with self-supervised and hierarchical deep learning. Earth and Space Science, 13(5), Article e2025EA004881. https://doi.org/10.1029/2025EA004881.

Moore, S. K., Dreyer, S. J., Ekstrom, J. A., Moore, K., Norman, K., Klinger, T., Allison, E. H., & Jardine, S. L. (2020). Harmful algal blooms and coastal communities: Socioeconomic impacts and actions taken to cope with the 2015 U.S. West Coast domoic acid event. Harmful Algae, 96, Article 101799. https://doi.org/10.1016/j.hal.2020.101799↗.

NASA Ocean Biology Processing Group. (2025 a.). Aqua MODIS Level-3 Global Binned Remote-Sensing Reflectance (RRS) Data, version 2022.0 [Dataset]. NASA Ocean Biology Distributed Active Archive Center. https://doi.org/10.5067/AQUA/MODIS/L3B/RRS/2022.0 Date Accessed: 2026-07-16

NASA Ocean Biology Processing Group. (2025 b.). Sentinel-3A OLCI Level-3 Global Mapped Earth-observation Reduced Resolution (ERR) Remote-Sensing Reflectance (RRS) Data, version 2022.0 [Dataset]. NASA Ocean Biology Distributed Active Archive Center. https://doi.org/10.5067/S3A/OLCI/L3M/ERR/RRS/2022.0 Date Accessed: 2026-07-16

NASA Ocean Biology Processing Group. (2025 c.). Sentinel-3B OLCI Level-3 Global Mapped Earth-observation Reduced Resolution (ERR) Remote-Sensing Reflectance (RRS) Data, version 2022.0 [Dataset]. NASA Ocean Biology Distributed Active Archive Center. https://doi.org/10.5067/S3B/OLCI/L3M/ERR/RRS/2022.0 Date Accessed: 2026-07-16

NASA Ocean Biology Processing Group. (2025 d.). Suomi-NPP VIIRS Level-3 Global Binned Remote-Sensing Reflectance (RRS) Data, version 2022.0 [Dataset]. NASA Ocean Biology Distributed Active Archive Center. https://doi.org/10.5067/SUOMI-NPP/VIIRS/L3B/RRS/2022.0 Date Accessed: 2026-07-16

NASA Ocean Biology Processing Group. (2026). PACE OCI Level-3 Global Mapped Apparent Optical Properties Data, version 3.2 [Dataset]. NASA Ocean Biology Distributed Active Archive Center. https://doi.org/10.5067/PACE/OCI/L3M/OC_AOP/3.2 Date Accessed: 2026-07-16

National Oceanic and Atmospheric Administration. (n.d.). Economics and demographics. NOAA Office for Coastal Management. Retrieved June 17, 2026, from https://coast.noaa.gov/states/fast-facts/economics-and-demographics.html ↗ .

Natural Earth. (2009). Ocean (Version 5.1.1) [Data set]. https://www.naturalearthdata.com/downloads/10m-physical-vectors/10m-ocean/ ↗.

Perry, R. I., Nemcek, N., Hennekes, M., Sastri, A., Ross, A. R., Shannon, H., & Shartau, R. B. (2023). Domoic acid in Canadian Pacific waters, from 2016 to 2021, and relationships with physical and chemical conditions. Harmful Algae, 129, 102530.

Trainer, V. L., Kudela, R.M., Hunter, M.V., Adams, N.G., & McCabe, R. M. (2020). Climate extreme seeds a new domoic acid hotspot on the US West Coast. Frontiers in Climate, 2, Article 571836. https://doi.org/10.3389/fclim.2020.571836.

U.S. National Office for Harmful Algal Blooms. (n. d.). Pseudo-nitzschia. Retrieved June 17, 2026, from https://hab.whoi.edu/species/species-by-name/pseudo-nitzschia/.